



URBADAPT-HEAT v1.0 — Urban Heat Adaptation Framework

Probabilistic, city-level cost–benefit analysis of heat-mortality adaptation



City-agnostic `template/` driven by `configs/<city>.yaml`



`data_manifests/`

Per-city manifests pointing to every source dataset, synced from Drive at setup

Inputs & grid harmonisation

NB1 Setup & Data Sync

01_setup_0126.ipynb

Load & validate city YAML · resolve paths
Sync inputs from Drive manifest
Input audit: FUA, LCZ, GVI, historical UrbClim
T2M archive, GVI–LST cooling coefficients

→ Validated config, resolved paths, masked LCZ raster

NB2 Harmonised Grids

HAZARD · EXPOSURE · VULNERABILITY

02_grids_0126.ipynb

Reference grid + FUA mask (100 m, EPSG:3035)
Hazard fields: synthetic baseline year (month/day climatology of the historical archive) + monthly CMIP6 warming deltas → daily T2M for each future target year
Exposure: WorldPop age-structured population (<15 · 15–64 · 65+), conservative reprojection; WCDE age-specific scaling for future years
Vulnerability: baseline SVI — thermal (GHS-AGE vintage) + foreign-born share + non-employment, equally weighted

→ Daily T2M files · age grids · baseline SVI

Climate risk and impact modelling

NB3 Hazard, Exposure & Dynamic Vulnerability

03_hazard_exposure_0126.ipynb

CLIMADA Hazard (T2M , °C, one event per calendar day × target year; frequencies corrected to sum to one per year)
Age-differentiated Exposures (one `impf_T2M` per age group)
Projected SVI per year & scenario (persistence k , spatial retention ϕ , thermal-retrofit pathway) → `exposure_with_vulnerability` read by NB04–NB08

→ Hazard + exposure + time-consistent vulnerability

NB4 Impact Functions

04_impact_functions_sensitivity.ipynb

Main: Masselot city- *and* age-specific ERFs — genuinely local, not a European average. `IF_MAIN_FAMILY` selects the high- T tail: constant vs **log-linear** (default)
Sensitivity: Burke polynomial & convex power-law (city-agnostic); cold-city variant for Track B
Annual scaling by projected baseline mortality (WCDE life tables)

→ Per-year `ImpactFuncSet` (main + sensitivity families)



Age-resolved heat-attributable mortality — current-AC baseline (no additional policy)

Adaptation options: effectiveness, costs, synergies, and trade-offs

NB5 Air Conditioning

PRIVATE

05_AC_0126.ipynb

NUTS3 penetration → zone-level coverage via a **logistic income-rank curve**, calibrated to reproduce the NUTS3 mean (observed sub-municipal income or `income emulator`)
Two policies under an **equal-users constraint**: income-targeted catch-up vs uniform uplift
Age-specific efficacy ϵ_a attenuates the mortality dose
Costs: CAPEX + maintenance + electricity (kWh/user dispersed by local summer heat exposure)

→ Impact function ↓

dose scaled by $(1 - \epsilon_a \cdot c)$

⚡ **Externality:** waste heat → hazard ↑
Empirical penetration → night- ΔT lookup + temperature-dependent COP degradation, CDD-share activation

→ GROSS avoided deaths · costs · marginal deaths per °C

NB6 Early Warning

PUBLIC

06_EWS_0126.ipynb

Warning days triggered by a **baseline-deaths threshold** (`epi_deaths_file`, periodically recalibrated; `event_mask` fallback where heat deaths are too few — Track B)
Efficacy interpretation follows a city-specific **three-tier maturity taxonomy**:
marginal — mature epidemiological HHWS with surveillance (e.g. Rome, Lisbon); added protection only
intermediate — meteorologically triggered HHWS (e.g. Berlin, Vienna); midpoint of marginal & counterfactual efficacy
counterfactual — no comprehensive HHAP (e.g. Athens, Copenhagen); deployment from a low baseline
Setup CAPEX graded by tier — zero for mature systems
Institutional ramp-up · gross → net via mortality displacement
Costs: per-capita per-warning-day (default) or per-day city-level model

→ Conditional mortality ↓

on warning days only

⇔ **Overlap:** efficacy scaled down by population-weighted current-AC coverage — AC shrinks the addressable residual mortality pool

→ Avoided deaths · costs over the CBA horizon

NB7 Urban Street Trees

PUBLIC

07_vegetation_0126_emulator2.ipynb

Baseline greenness from the **Green View Index**, by LCZ class
Default policy: **regional Q3 catch-up** — every region below the city-wide upper quartile receives its ΔGVI shortfall, built LCZ classes only
Two-step cooling: $\Delta GVI \rightarrow \Delta LST$ from city/month/LCZ regressions (cooling-only, warming clipped) then $\Delta LST \rightarrow \Delta T2M$ via a `quadratic emulator`
Maturity ramp · asset lifetime · equity-weighted allocation variant
Costs: city-specific PLI-adjusted CAPEX + O&M per tree

→ Hazard ↓

$\Delta GVI \rightarrow \Delta LST \rightarrow$ `quadratic emulator` $\rightarrow \Delta T2M$

🌳 **Co-benefit:** cooling → AC electricity ↓ at a configured reduction rate per GVI point

→ Avoided deaths · costs · equity metrics



Policy-specific incremental costs, avoided deaths & interaction parameters

Economics, uncertainty & synthesis

NB8 Cost–Benefit Analysis

08_CBA_0126.ipynb

Integrates NB5 + NB6 + NB7 on a common discounted horizon; hazard track read from `HAZARD_TRACK`
Anchor-year mortality interpolated to annual streams; trees scaled by maturity; gross → net displacement; optional life-years saved

Explicit bookkeeping instead of silent netting:

`AC (NET)` — standalone, minus standalone waste-heat penalty

`AC (NET + trees)` — combined: lower kWh/user, λ_γ -corrected penalty

`Trees (base O&M)` — direct health benefits only

Pareto frontier over policy scales + ICERs along it; benefits by SVI quintile; public vs private cost bearer

→ PV cost, NPV, BCR, cost/death, cost/life-year, ICER, efficient portfolios

NB9 Uncertainty Quantification

09_uncertainty_0126_improved_fast.ipynb

→ cityheat.nb09_improved_fast

Monte Carlo propagation through the full chain over: climate scenario & GCM band (low / central / high)
IF family, tail extrapolation & T_{ref}
AC / EWS / tree efficacy & maturity timing
SVI projection (k , ϕ , retrofit rate)
costs, cost model choice & discount rate
Spatially explicit **daily raster** modifiers for tree cooling and AC waste heat; event-level warning-day logic
PAWN global sensitivity indices per output

→ Probabilistic deaths avoided, BCR distributions & factor ranking

NB10 Summary & Reporting

10_summary_0126.ipynb

→ cityheat.nb10_summary

Aggregated results tables & figures across pathways, scenarios and hazard tracks
Ranking of adaptation options by cost-effectiveness and equity
Output manifests record every configuration choice, so a run is reproducible from config alone
Variant-namespaced, self-contained per-city report

→ Decision-ready comparison, reproducible per city & variant

Inputs & grids

Risk model

Adaptation

Economics & uncertainty

Statistical emulator